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An Experimental and Comparative study about the engine Emissions of conventional Diesel Engine and Dual Fuel engine

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Abstract:

Because of the high thermal effectiveness, consistency, flexibility and economical cost diesel engines are extensively used all around the world. Diesel engine emissions are producing serious environmental pollution which consists of oxides of nitrogen (NO_X), carbon monoxide (CO) and particulate matter (PM). So it's necessary to find the alternate solution to control diesel emissions. Natural gas is a highly attractive due to its clean burning, low cost, and wide availability. In this experimental work, single cylinder (8.6 Hp LA186F) four-stroke conventional diesel engine was studied. Natural gas is the major gaseous fuel used in dual fuel method which is up to 80% while diesel is used as a pilot fuel for the source of ignition up to 20%. In comparison with the conventional diesel method, it was observed that the dual fuel method significantly reduces the NO_X maximum up to 72.5%, carbon dioxide (CO_2) 34.5% and CO 59.8%. However, Hydrocarbon (HC) increased 66.76% in contrast with normal diesel combustion. During dual fuel mode, the emissions of HC and NO_X shows the inverse relation.

Keywords: *Diesel; Dual fuel; Emission; Natural gas; Combustion.*

1. Introduction

Diesel engines are extensively used in the public transportation because of their higher stability and thermal effectiveness. Meanwhile, greenhouse gases are produced by transportation sector is about 30% of the world which leads to global warming [1]. Therefore, diesel engine is more responsible for the serious atmospheric hazards [2]. NO_X and PM are the main harmful components of diesel engine. NO_X emission produces photochemical smog which is a major source of acid rain [3]. It has been proved by numerous studies that these gases can be enormously harmful to the environment, especially living beings as health hazard [4]. Hence, emission regulations become stricter to reduce this environmental pollution. At the same time, energy consumption is growing rapidly and the fossil fuels are dwindling. In this regard, use of alternative fuels is necessary to meet the requirement of energy consumption and standard emission regulations. In public transport, natural gas is more environmental friendly among the various alternative fuels. Because of its clean burning, widely spread distribution stations, and lower cost [5].

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